

Enhancing Cyclist Safety Through Driver Gaze Analysis at Intersections With Cycle Lanes

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Abstract—In urban areas, roads with dedicated cycle lanes play a vital role in cyclist safety. However, accidents can still occur when vehicles cross the cycle lane at intersections. Accidents mostly occur due to failure of the driver to see a cyclist on the cycle lane, particularly when the cyclist is going straight through the intersection, and the vehicle is turning. For safe driving, it is critical that the drivers visually scan the area in the vicinity of the junction and the car, particularly using the wing-mirror, prior to making turns. This paper describes results from a set of test drives using in-vehicle non-invasive eye-tracking and in-vehicle CAN bus sensors to determine driver behaviour. In total, 20 drivers were monitored through 5 different intersections with cycle lanes. The study found that approximately 83% of drivers did not check their wing mirror prior to, or during their turning manoeuvre, potentially putting pedestrian, cyclists, scooter and hoverboard users in danger. An algorithm was developed to analyse driver gaze during the turning manoeuvre to identify cases where they failed to look at the wing mirror. The gaze pattern and gaze concentration on the mirror helps to identify safe and unsafe driving behaviour. This information can then be used to improve Advanced Driver-Assistance Systems (ADAS) to create a safer environment for all road users.

Index Terms—Driver inattention, mirror checking detection, ADAS, driver gaze behavior, road safety, situational awareness, nearside-turns.

I. INTRODUCTION

DRIVERS are required to scan the periphery of their vehicle to keep up to date with the dynamics of surrounding traffic. This requirement becomes critical when a vehicle is turning. Most modern vehicles are equipped with sensors that support drivers to help them see areas around the vehicle they cannot view directly e.g. reversing cameras allow drivers to view the area immediately behind their vehicle. Driving tasks

rely heavily on visual information collected by the driver [1]. Normal visual scanning during driving requires drivers to check both wing-mirrors as well as the rear-view mirror [2]. When a vehicle is turning at an intersection with an adjacent cycle lane, good driving behaviour becomes critical for road safety. Missing out on even a short glance at a side-view mirror prior to turning can lead to compromised situational awareness and could put cycle lane users in danger.

In many countries, drivers are not only required to check their mirrors but are also taught to turn and look over their shoulder before making a turn. This practice is considered a crucial safety measure, and failing to do so can result in failing the driving exam. However, turning the head while driving can be a dangerous activity, as it momentarily takes the driver's attention away from the front of the vehicle. To compensate, many drivers rely on looking intently into the mirror and moving their head to pan their view, which is also time-consuming and diverts attention from the road ahead but keeps the front of the vehicle in their peripheral vision.

Cyclists are vulnerable road users in comparison to motorised vehicles and are at higher risk while sharing the same road space as motorised vehicles. In 2019, approximately 2000 cyclists died in road accidents in the EU, accounting for more than 9% of total EU road death, and this figure is increasing [3]. However, the actual proportion is likely much higher due to the underreporting of these crashes [4]. Due to increased traffic congestion in urban areas, and to help to reduce carbon footprint, more people are moving towards the use of bicycles and e-scooters. Although the number of all road fatalities decreased in the EU, the percentage of cyclist fatalities compared to all road fatalities saw an increase from 6% in 2007 to 8% in 2016 [3].

To facilitate the growing number of cyclists, many improvements have been made to urban road structures such as the inclusion of a separate cycle lane near main roads and separate cycle tracks. A survey conducted in Dublin, Ireland in 2012 found that 57% of cyclist said a more connected on-road cycle lane system encouraged them to commute via bicycle [5].

Although cycle lanes and tracks have improved cycling facilities, more accidents occur at intersections with cycle lane/track and motorised traffic than any other section of the road [6], [7], [8]. In situations where both cyclists and vehicles are moving, the chances of a collision between a cyclist and vehicle are between 2% to 8%, depending on the layout of the intersection [9]. Accidents at intersections

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are a major contributor to the overall increase in bicycle accidents and a large percentage of these occur at signalized intersections [10]. In the EU, cyclist accidents at junctions account for 25% of all road accidents at junctions [3]. In EU countries, 28% of cyclist fatalities occurred at junctions [11]. Jansen [10], reported an increase in the number of injuries to cyclist and moped riders at intersections with on-road cycle lanes. According to the European Road Safety Observatory report in 2018, 58% of cyclist fatalities occurred in urban areas [11]. Miranda-Moreno et al. reported that in Canada a 10% increase in bicycle users resulted in a 4.4% increase in cyclist related injuries while a 10% increase in vehicles passing through intersections resulted in a 3.3% increase in cyclist-related injuries [12]. Studies also showed that nearside turn maneuvers (right turn in the EU and USA and left turn in UK and Ireland) have a significant impact on the rate of cyclist-related injuries [12]. A study conducted in Denmark by the Danish police department showed that 16.5% of cycling injuries between 2012 and 2014 involved a right-turn (nearside) movement [13]. Wu and Xu reported that drivers may pay less attention to their right, where relevant VRUs might approach, particularly when these VRUs have the right-of-way while the driver is completing a right turn on red (RTOR) [14].

E-scooters are a recently introduced mode of transport that are growing in popularity, particularly in urban settings. As yet, there is no dedicated infrastructure for e-scooters and in many countries there are few rules or regulations applied to e-scooters hence e-scooter riders frequently use either pedestrian or cycle lanes, effectively incorporating them into same category as cyclists in a road safety paradigm. The use of pedestrian infrastructure due to absence of legislation puts e-scooter riders at greater risk particularly when they look like a static pedestrian, yet potentially move with the speed of a fast cyclist. For example an amendment AB-2989 in California law allows e-scooter users to ride without a helmet and can speed up-to 35 km/hr [15]. Many cities and councils have been forced to introduce ad-hoc policies in the absence of more regulated national policies [16]. E-scooter rental schemes in urban areas have also increased the number of non-regular users. These occasional users are often not experienced and certainly not trained in dealing with traffic while riding e-scooters, and are, therefore, likely to be prone to making more mistakes than regular users. Although an e-scooter may be a cheaper alternative to buses or taxis, the increase in number of accidents raises concerns [15], [16]. Bai et al. concluded that e-scooter users tend to engage in more risky behaviour at intersections than cyclists and e-bikers [17]. E-scooter users have been found to bypass signalized intersection more aggressively than bicycles or e-bikes [17]. In a study in China, Gua et al. found that 25% of e-scooters run red lights at signalized intersections compared to 18% of bicycles [18]. With a lack of infrastructure, knowledge and legislation, the burden of road user safety often lies with car drivers at a signalized intersection.

Given these risks, it is critical to understand and assess drivers' mirror-checking behavior before turning. While checking mirrors is considered one of the primary driving tasks [19], [20], [21], the combination of mirror checks

and shoulder checks (looking outside of driver-side window) provides a more comprehensive safety approach. Which is why good driver behaviour is associated with checking your surrounding using mirrors, particularly when taking turns or changing lanes [20]. According to the National Highway Traffic Safety Administration (NHTSA), drivers must check mirrors prior to initializing turning maneuvers [22]. Road accidents involving cyclists are often linked to either a lack of driver awareness or driver misjudgment, such as the driver failing to observe the true location and/or speed of the cycle lane user, leading to 18% of all cyclist fatalities [23]. Large vehicles often have many blind-zones where cyclists effectively disappear from the driver's vision. The term blind-zones is used instead of blind-spot as it describes an area around a vehicle rather than a specific spot. Vehicle blind-zones can be described as "areas around the vehicle that cannot be seen directly by the driver by looking forwards or by using any of the vehicle's standard rear-view mirrors (internal and external) from the normal sitting position" [24]. Dooley et al. showed that a blind-zone could be as large as 10m long and 3m wide [25]. One solution to this problem has been to increase the number of mirrors around a vehicle [26], [27]. Unfortunately when taking nearside turns at a junction where a cycle lane is also present, drivers often fail to check the nearside wing mirror which may result in their not seeing a cyclist [19], [28].

The number of collisions between cyclists and vehicles can be significantly reduced if an action is taken on the knowledge of a drivers mirror checking behaviour prior to taking a turn. Of course this is no guarantee that the driver has seen the cyclist or that they will act appropriately, but it is indicative of two things: (1) the driver is alert and attentive and (2) the driver has looked in a region where an at-risk cyclist may be located. Several techniques have been investigated to study driver mirror checking action. The most common techniques are head pose tracking and gaze tracking. The most reliable sensors for this task are inward facing cameras [21], [29], [30], [31], [32] and eye trackers [19], [33], [34], [35]. Driver head pose can be used to estimate driver gaze direction, however, eye tracking offers more reliability [19].

The most significant challenge in determining driver situational awareness is that there is no way to objectively determine or quantify the level of awareness of the driver. The most cited model of situational awareness is Endsley's model of Situation Awareness [36]. This model describes three levels of situation awareness: i) perception, ii) comprehension and iii) projection. Without perception the other two levels are nullified. Hazard perception is the ability of a driver to perceive a potentially dangerous situation on the road ahead. Lack of hazard perception leads to compromised situational awareness and hence, is closely linked with driver-related accidents [37], [38], [39]. Crundall et al. found that driver perception is closely linked to the direction of the eyes [40], however it is impossible to determine whether drivers have registered the information they have perceived from a scene they have recently looked at. It is easier to determine driver inattention in real time by identifying lack of perception than trying to determining level of awareness [41].

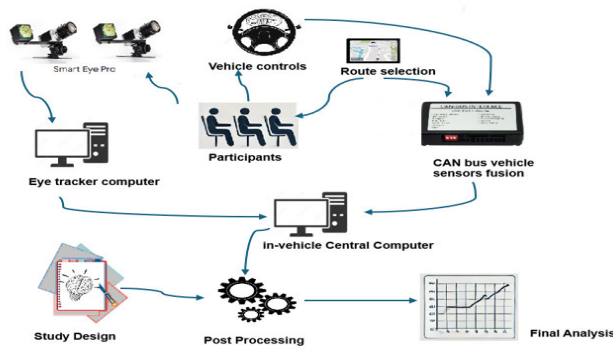


Fig. 1. Data Flow Diagram of the methodology adopted in this study.

II. METHODOLOGY

Signalized intersections with cycle lanes were chosen in the city of Galway, Ireland, to study driver gaze behaviour during a left turn. An approach to represent the driver gaze direction as a vector in a 3-D space has been utilised by Christer et al. and the preceding study of this experiment by Abbasi et al. [19], [33], [41]. Driving-related static areas of interest are defined within the vehicle and driver gaze direction is studied with respect to those areas. Areas of interest are defined by the manoeuvres, from simply driving straight ahead, to lane changing, left or right turn and roundabouts. Xucong et al. [42] used head pose data to estimated the gaze as a point in 3-D space. Vicente et al. [43] estimated driver head pose and gaze off the road using a camera with an IR illuminator. Franco et al created space in 3-D and labelled it as a primary attention zone, comprising of the central part of the frontal windshield, and in 3d space determined whether drivers gaze fell in or out of the primary attention zone. Gazes falling on the attention zone are labelled as attentive responses while gaze falling outside of the zone are labelled inattentive or away gazes.

Figure 2 describes a typical urban intersection including a cycle lane. In this type of intersection, a cycle lane runs parallel to the left turning lane. Figure 2 also shows the key points of area under observation for this study. Point A indicates the starting point of the observation which is 50 meters from the stop line. The regions between points B and C indicates the last 20 meters before the stop line when the segregation is suspended because both vehicles and cyclists share the junction [45]. In the scenario under investigation when the traffic light turns green, the vehicle in the left lane makes a left turn while a cyclist proceeds in a straight line. In this situation, mirror checking action becomes one of the most important indicators of a driver's ability to perceive potential danger in the cycle lane. Figure 1 illustrates the key elements of the data flow structure employed in the methodology for this study.

A. Participants

A cohort of 24 drivers were recruited to take part in this study. The recruitment process included drivers with a minimum of 2 years clean driving experience and a full driving license. The age profile of the participants was between 22 and 77 years with the majority being between 22 and 40. Drivers

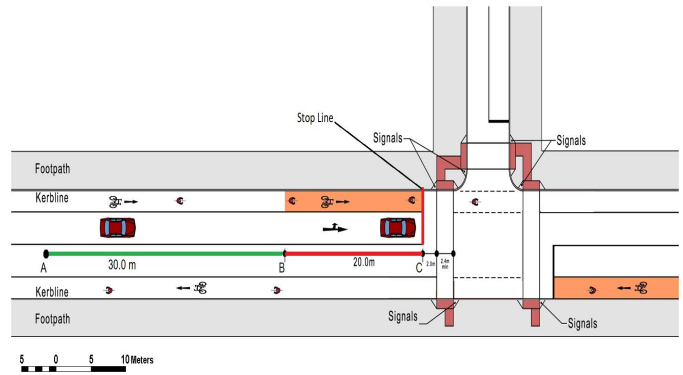


Fig. 2. A typical urban intersection with an adjacent cycle lane. The orange colour represents the last 20m of the cycle lane before the stop line. (Adapted from a figure courtesy National Transport Authority, Ireland.)



Fig. 3. SmartEye tracker setup Cockpit view of the test vehicle illustrating the placement of eye trackers and illuminators in the vehicle dashboard, and a scene camera placed behind the driver's head.

requiring spectacles while driving were excluded due to the difficulties in determining their gaze with the IR eye tracking system. This study received ethical approval by the National University of Ireland, Galway and all participants provided consent for this study.

B. Vehicle Sensors

To monitor driver gaze, a SmartEye Pro [46] eye-tracking system was fitted in a Citroen C4 Picasso. natural gaze behavior. To ensure comprehensive eye tracking coverage, the SmartEye Pro system consisted of three sensors mounted on the car dashboard as shown in Fig. 3. Two IR illuminators were installed to illuminate the driver's retina. The system was calibrated for each participant before conducting each test drive. The system is capable of capturing up to a 180-degree range of head movement. This setup enabled robust eye tracking even during head turning and tilting movements, providing a wider field of view for analysis

The vehicle was equipped with a CAN bus which carries information from all vehicle sensors and systems including accelerator pedal pressure, vehicle speed, steering wheel angle and brake pressure. CAN bus data were recorded and time-stamped by a computer in the vehicle. A GPS system was also used to identify the location of intersections.

C. Study Design

Driver monitoring studies are typically carried out in controlled environments or simulated settings, where specific

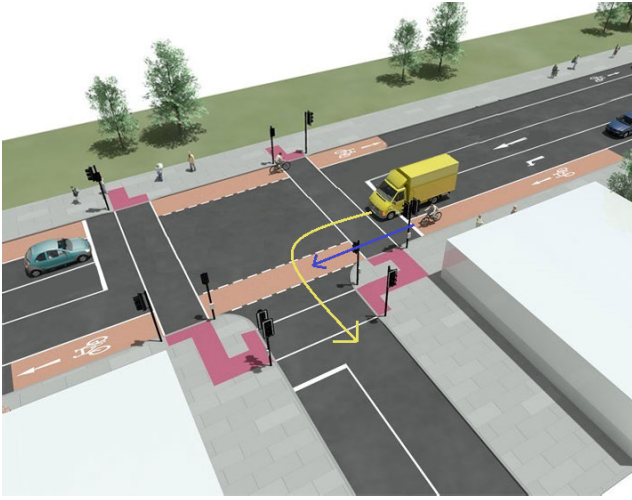


Fig. 4. A dangerous scenario where a yellow van is about to make a left turn (indicated by the yellow arrow) while a cyclist intends to proceed straight ahead (indicated by blue arrow). (photo adapted from a figure courtesy National Transport Authority, Ireland.)

objects or scenarios are introduced to manipulate the driver's gaze. However, Otto Lappi et al. [44], introduced a methodology that involved naturalistic on-road driving in wooded areas, without providing any explicit instructions or tasks to the driver. The study specifically examined the driver's gaze behavior and their level of inattention towards the road while navigating curves.

This study was conducted in real on-road traffic conditions and the drivers were not given any specific instructions or tasks, allowing for an examination of their natural gaze behavior. The test route consisted of multiple driving scenarios from urban, rural, sub-urban to dual carriageway. Intersections in urban traffic were studied to determine driver gaze behaviour on the left-view mirror to check the cycle lane users. In total 5 intersections were chosen with dedicated cycle lanes. A typical example of an intersection with a cycle lane is shown in Figure 4. Drivers were not informed about the cycle lane test at intersections so that data remained unbiased throughout.

Specific areas within the vehicle were defined as gaze targets, [41], to study drivers visual behaviour while taking a turn. These gaze targets included the left wing mirror, side windows and the frontal area (a subset of the windshield immediately in front of the driver). To scan the rear and side periphery of the vehicle, drivers use wing and rear-view mirrors. The gaze targets are specified virtually in a 3D world model of the eye tracker software using the world coordinate system (WCS). Using the WCS, the system can precisely determine which real-world location inside the vehicle the driver gaze is intersecting with. The gaze vector is defined as the direction of the gaze, together with the original point in 3D space. During calibration, separate gaze measurements are taken for the left and right eyes that are later combined into a consensus gaze measurement to extract true eye position in the virtual world. During each test drive, driver glances and their durations are recorded in the system as a gaze target time series.

D. Stopping Time and Distance to Stop Line

It is not possible to objectively measure driver attention and intention during, or before making a decision to take a

turn. To alert the driver or to effectively deploy an emergency system sufficiently early to prevent a collision, the speed of the vehicle must be known in order to calculate the vehicle stopping time and distance. Stopping time is also linked with driver reaction time, weather, road condition and brake efficacy which in turn affects the choice of starting point of our drivers left mirror checking analysis. Points A, B and C are shown in Figure 2 with point A being the starting point of driver mirror checking action analysis. Point C indicates the stop line of the intersection. Point B is taken as 20m before the stop line where the cycle lane merges with the road as shown in orange in Figure 1. The speed limit on Irish urban roads is 50 km/h. This study found that the average speed of vehicles approaching a left-turn is approximately 25 km/h (the average speed taken between point A and point C). The typical braking distance of a car travelling at 30 km/h is approximately 11m in dry and 15m in wet conditions [47]. The 50m distance to the stop line was chosen to accommodate all weather conditions and could be altered to suit a different scenario, or to accommodate different traffic speeds. Fambro et al. reported that driver perception-braking response time in an unexpected scenario is about 1.1 seconds [48]. The vehicle can cover up to 7 meters in 1.1 seconds when travelling with the speed of 25Km/h. At higher speeds, the stopping distance increases.

III. RESULTS AND ANALYSIS

In the context of driving, what a driver perceives is inseparable from where they are looking [50], because humans tend to turn their gaze towards to a specific object or location to acquire detailed information from a scene. As driving is primarily a visual task [1], analyzing the direction of a driver's eyes and their movement can contribute to assessment of driver attention or, lack of attention. If a driver has failed to look at a specific area, for example rear view mirror while reversing the vehicle, it is reasonable to assume that the driver has missed the information contained in that area. Analysis of the direction and movement of the eyes can contribute to the assessment of inattention [51]. In the following section, driver gaze behaviour and other factors such as speed of the vehicle and road infrastructure are analysed for the safety of the occupants of the vehicle and cycle lane users.

A. Speed of Approach

The average speed of approach to all five signalized intersections evaluated in this study was recorded from point A to point C as per Figure 2. Vehicles approaching on a red signal were treated similarly to those approaching on green. The variation in approach speed was influenced by factors such as traffic density, the vehicle's position in the queue, and whether the vehicle was approaching a stale green light. The speed metric reflects only the average speed of the vehicle while passing between points A and C, that is, entering the studied section and crossing the stop line. A vehicle may come to a complete stop at any point between A and C and then accelerate again to pass point C. The average speed for different junctions varied from 15 km/h to 30 km/h depending on the traffic density. The average speeds of traffic approaching all intersections is shown in Figure 5. The highest speed is

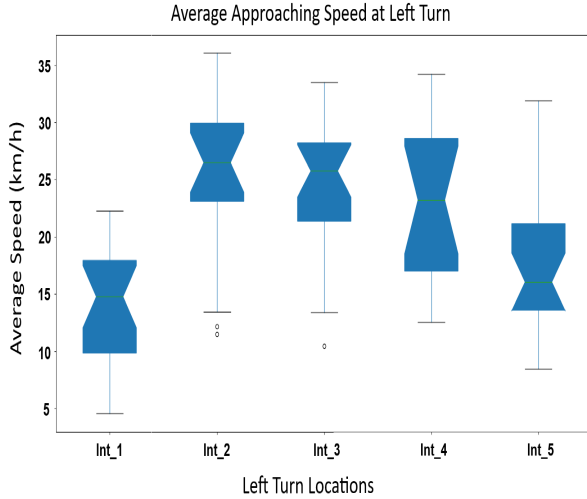


Fig. 5. Average speed of all left turns illustrating the average approaching speeds of all vehicles on intersections with cycle lanes.

measured when the lane ahead is clear of other traffic and drivers can approach the junction at maximum permissible speed. Cycle lane users such as e-scooters, e-bikers and cyclists can reach speeds of up to 30Km/h while using the cycle lane. When the cycle lane user matches the speed of an adjacent car, it could potentially meet a turning car at the apex of the intersection.

B. Eyes on the Targets

A Left turning car's most important gaze targets would be its left wing mirror, left window and the frontal area. To determine whether a driver has looked at the desired gaze target, it is first important to identify the minimum glance time required by the human eye to register information from a scene. The minimum glance time is taken in this study as a reckonable period for a 'look' to be registered. The basic parameter used to study human gaze behaviour is fixation [52], [53]. Fixations are the very brief pauses of the eyes during which the human brain extracts visual information from a scene. The ISO 15007-1:2014 standard [54] defines that fixation time varies between 100ms and 2000ms. Mackeben et al. also support a minimum fixation time by suggesting that a human cannot see an object in its foveal vision if it disappears within 200ms of appearing in its view [55], [56]. Therefore, 200ms has been used in this paper as the minimum required fixation time. It is, however, not possible to prove that drivers have seen and understood the relevant information from a scene. This minimum fixation time is also supported by major remote eye trackers manufacturers such as SmartEye (the tracker used in this study) and Tobii. Raw data from the eye trackers were processed using a median filter to remove any glances less than 200ms in duration.

Figure 6 shows driver gaze behaviour on left wing mirror at the 5 different intersections studied in this research project. As can be seen in Figure 6, the majority of drivers did not look at the left-view mirror prior to taking the left turn. It was found that from starting point A to the stop line at point C (Figure 2), 85% of drivers did not register a single fixation on the left-wing mirror. Simply put, 85% of drivers could have

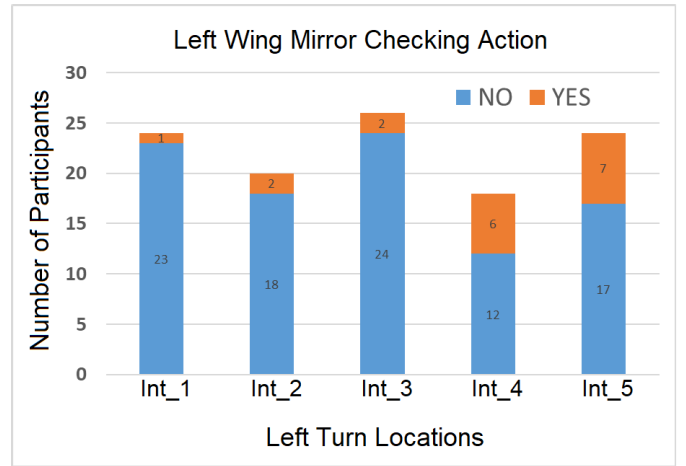


Fig. 6. The number of drivers at each intersection who looked at the left wing mirror before turning.

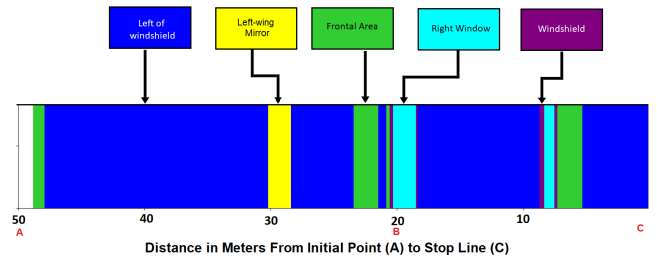


Fig. 7. Driver gaze behaviour illustrating driver gaze behaviour on left mirror and other gaze targets from the start point to stop line before making a left turn.

collided with a cyclist in the cycle lane because they did not check their mirrors prior to making the turn.

C. Minimum Visual Attention Requirement

Drivers share their visual attention between the centre of the road and other visual targets or locations during routine driving [33]. This visual time-sharing is defined by the distribution of glances between different scenes to gather the information from each scene to be able to generate the gist of a complete situation around the vehicle [57]. Generally, when moving forward, drivers do not look away from the centre of the road for more than 1.5 seconds [58]. If drivers do not capture the visual information required in one glance at a secondary target, they frequently shift their gaze between the frontal area and the other gaze target [50]. According to NHTSA recommendations a driver should not take their eyes away from the centre of the road for more than two seconds [59].

Figure 7 depicts a representative instance of a driver directing their gaze towards specific targets while progressing from point A to C in preparation for a left turn. As observed in Figure 7, the driver focuses their gaze on the left side of the windshield area while driving, and additionally, briefly shifts their attention to the left-view mirror in anticipation of the left turn maneuver. A safe driver's appropriate gaze behavior should encompass at least one fixation on the left-wing mirror before initiating the left turn, as exemplified by this particular driver.

Figure 8 shows the glance distribution of a driver who looked at the left-view mirror and left windows at all five junctions under investigation. It is clear that drivers intuitively

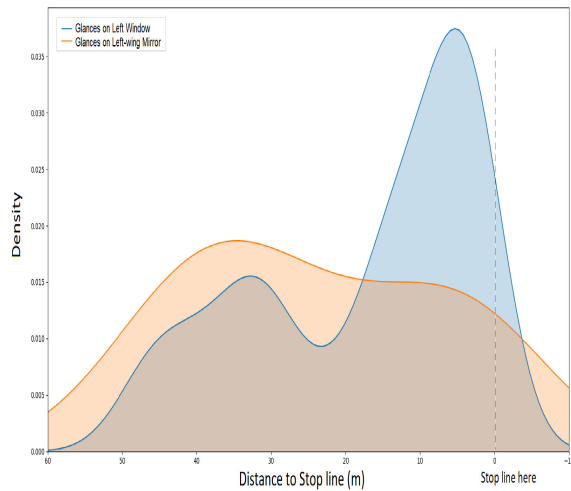


Fig. 8. Glances distribution with respect to distance to stop line in meters.

look through the left window and left-wing mirror just before taking the turn. It can be inferred that due to turning of the vehicle the head movement of the driver is naturally tilted towards the direction of the turn which resulted in more glances falling on the left window in the final 10 meters on the approach to the stop line. However, looking through the left window can only provide information about a vulnerable road user (VRU) beside, or slightly in front of the vehicle on the cycle lane. A problem arises when a cyclist or e-scooter user on the cycle lane approaches faster than the vehicle, and tries to overtake the slower moving, turning vehicle. Considering driver reaction time and the time it takes to safely stop a moving vehicle, information gathered prior to 10 meters from the stop line will become important for drivers to be able to safely stop the vehicle in an emergency. It is important that drivers have at least one fixation on the left-wing mirror before initiating a turn. If the driver fails to see a cyclist approaching from behind, or if he/she sees it at the very last moment, it is more difficult for them to react in time to avoid a collision.

D. Discussion

The findings highlight critical aspects of driver behavior, particularly the role of visual attention in ensuring safety at intersections, especially when cycle lane users are involved. The analysis of approach speed and gaze patterns shows how various factors contribute to potential hazards. Drivers' approach speeds, which vary from 15 km/h to 30 km/h, are influenced by traffic and signals, but this creates risks for cyclists, who may be traveling at similar speeds. The danger is heightened when drivers fail to notice cyclists approaching from behind, especially during turns.

Eye-tracking data reveals that while drivers typically glance at the left-wing mirror and window before turning left, the effectiveness of this gaze behavior varies. Filtering out glances shorter than 200ms helps identify processed visual information, but it doesn't ensure full comprehension. This becomes particularly dangerous with fast-moving cyclists or e-scooters, as late detection in the final 10 meters before a turn significantly increases the risk of collisions.

The study also underscores the limitations of human attention, especially at busy intersections with multiple visual targets like mirrors, cyclists, and vehicles. Research indicates that drivers should avoid glancing away from the road for more than 1.5 to 2 seconds, but complex urban settings can stretch these limits. This demonstrates the need for an advanced system that can step in when needed.

Driver gaze direction is key to predicting their next maneuver, as drivers intuitively look at the space they plan to occupy. Understanding these natural behaviors is crucial for developing algorithms to detect driver inattention. Such systems must carefully balance alerting drivers without overwhelming them, as excessive or unnecessary warnings could increase cognitive load and create new risks. For instance, a system should alert the driver if they fail to see a fast-approaching cyclist but should avoid interfering if the driver has already noticed the cyclist. In critical situations, the system may need to assume control of the vehicle to prevent a collision.

Figure 4 shows a scenario in which a yellow van in slow moving traffic has indicated their intention to turn left while a cyclist has the right to continue straight, but faces a potentially dangerous situation because they are approaching from behind, at a higher speed than the vehicle. In scenarios like these, cyclists more frequently notice the vehicle intending to turn than drivers notice the cyclists, and as EU law requires [61], cyclists often expect drivers to yield to give cyclists priority at an intersection [62] without knowing whether or not the driver has seen them. This highlights the importance of systems that ensure both driver and cyclist safety.

E. Algorithm Design

An algorithm has been developed to automatically detect driver gaze on the left-view mirror as a vehicle approaches an intersection. An overview of the algorithm is shown in Fig. 9. The algorithm has 7 danger levels from 0 to 6, with 0 being no alert and 6 indicating an impending collision. The algorithm commences observing driver gaze 50m before the stop line of the junction. The danger starts at level 1 at 50 m and is increased by one every 10 m thereafter. If at any point the driver looks at the left-wing mirror, the algorithm reduces the danger level to zero unless the vehicle is within 20m of the stop line, in which case the algorithm will retain a danger level 2 as a precautionary measure.

The algorithm considers both the current speed and distance to the stop line. Measuring vehicle speed is especially critical between points B and C, as it directly impacts the safety of cycle lane users. However, in developing the algorithm to alert drivers who may not be situationally aware of VRUs in the cycle lane, it is crucial to consider driver gaze behavior before reaching these points. The driver gaze behavior monitoring began from point A, where the cycle lane runs parallel to the car lane, and continued until point B, where the cycle lane approaches a merger with the car lane. Driver gaze behavior was closely observed, particularly toward the left view mirror, to determine whether the driver scanned for VRUs in the adjacent cycle lane as they approached the final phase of the turn. The area between points B and C is critical for driver reaction, and a driver who has noticed the cycle lane user before point B would likely be more vigilant in checking the

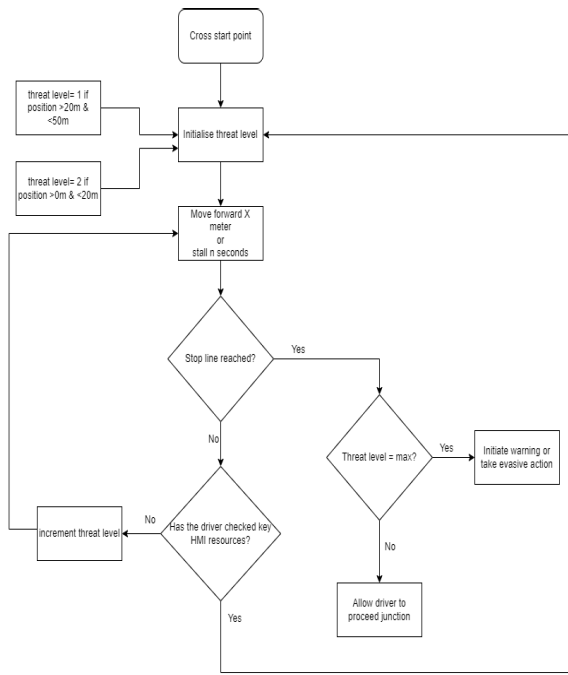


Fig. 9. Overview of driver monitoring algorithm at intersection.

left view mirror. Since it is challenging to confirm whether the driver has fully registered the presence of the cycle lane user, we can only monitor whether the driver has glanced toward the cycle lane using the left view mirror and window.

If the vehicle is stopped more than 20 meters from the stop lines (e.g. due to heavy traffic) then the algorithm sets the danger level to level 1, because drivers tend not to look for driving related information while stopped, and resume their scanning behaviour just prior to moving off again. As the vehicle is not moving, there is no danger of the vehicle colliding with a bicycle or other VRU. If the vehicle is moving faster, then the algorithm may increase quickly by 2 to 3 danger levels depending on the level of the danger calculated.

If the vehicle is moving slowly, the driver will spend more time travelling the final 20m, therefore the intervals between left-wing mirror checks become more important because the speed of the vehicle is close to the speed of a cyclist, potentially increasing the risk of collision between cyclist and vehicle. It is critical for drivers to look at the side-wing mirrors in the last 20m, especially when the vehicle is moving slowly. The danger level increases gradually and only reaches its maximum level 10m before the stop line, giving drivers enough time to react to avoid a potential collision. The system is designed to not only eliminate false alarms but also to give drivers time to bring their attention back to the left-wing mirror. It is customisable for a wide range of driving styles.

The danger level will remain at level 2 if the driver had a glance on the left-view mirror in the last 20m up to the stop line. A car with an average speed of 30km/h would cover the last 20m in little more than 2 seconds. It is reasonable to assume that the driver retains the information previously collected from the left-wing mirror. To allow the driver sufficient time to react to potential hazards, the threshold must be raised before the final 10 meters from the stop line,



Fig. 10. Cyclist approaching (yellow arrow) in the cycle lane within 20 meters of the car. The car is approximately 10 meters from the stop line.

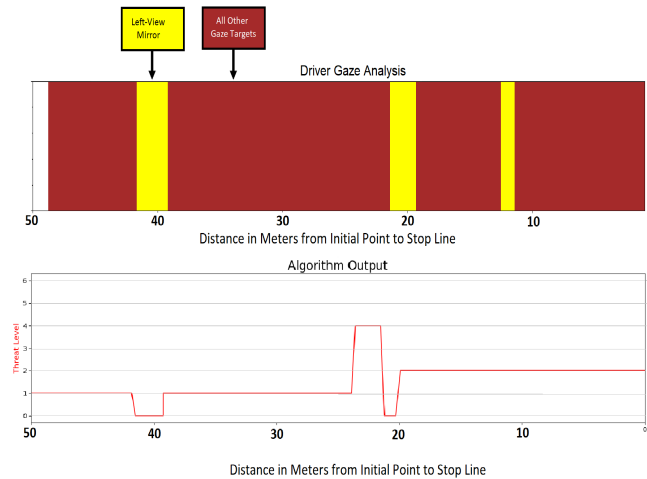


Fig. 11. Driver Left Mirror Interaction (safe driving) Illustrating the driver gaze falling on left mirror from the start point to stop line before making turn.

as the vehicle's speed between points B and C (illustrated in Figure 2) is critical for cyclist safety. This ensures there is enough time and space for the driver to take appropriate action if necessary. The algorithm only analyses driver gaze behaviour for a left turn and does not use any other sensors for any other potential road user in the cycle lane.

The algorithm has been tested on cases where the driver had frequent glances on the left-view mirror prior to taking a left turn. Figure 10 shows a scenario where a cyclist can be seen approaching on the cycle lane while the car is approximately 10 meters from the stop line and the traffic signal is green. A camera installed on the side-wing mirrors of the car recorded the view from the perspective of the side-wing mirror as the driver would see it and also covered the blind-spots around the vehicle. In this case, driver attention on the left wing mirror is vital to ensure that they have an opportunity to view the approaching cyclist.

Figure 11 shows an output from the algorithm for the case depicted in Figure 10 where it can be seen that the driver has regularly checked the left wing mirror prior to making a left turn. The driver's gaze allocation pattern on the left wing mirror depicts the type of safe driving behaviour required to execute a safe turning maneuver. In Figure 11, all other gaze targets are removed to only show driver glances on or off the left wing mirror.



Fig. 12. Cyclist on the road just about to pass the vehicle on the left while the car is crossing the stop line prior to turning left.

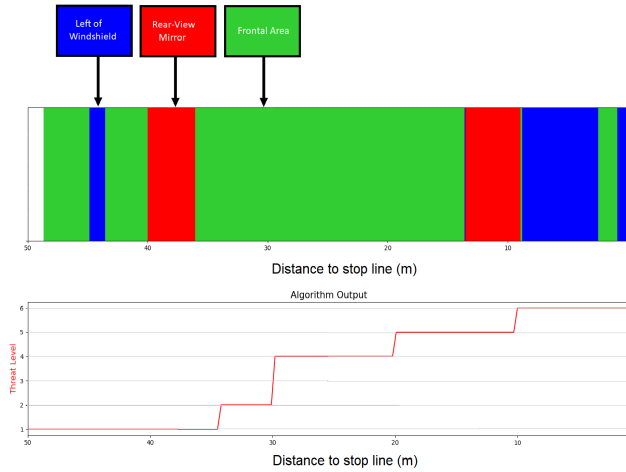


Fig. 13. Driver Left Mirror Interaction (unsafe driving) Illustrating the driver gaze falling on left mirror from the start point to stop line before making turn.

As can be seen from Figure 11 the driver looked at left wing mirror twice in the interval between 50m and 20m from the stop line. We call this the observation zone where driver gaze behaviour is observed and the danger level is updated accordingly, but no action is taken. The driver first glanced at the left wing mirror approximately 40m before the stop line. The combination of vehicle speed and lack of engagement with the left wing mirror raised the danger by two levels to bring it to level 4 approximately 20m before the stop line. The level was reset once the driver again looked at the left wing mirror. A glance at the left wing mirror in the last 20 meters again reset the danger to level 2 as a precautionary measure due to the proximity to the turn.

Figure 12 shows a situation where a cyclist came close to hitting the vehicle as it was about to take a left turn. The vehicle speed (extracted from the CAN bus) was approximately 20Km/h. Video data showed that the cyclist was present in the cycle lane beside the car. Eye tracking data revealed that the driver only shared glances between the front of the windshield and rear-view mirror prior to taking the turn. The driver never glanced at the left wing mirror, despite the fact that they turned on their left-turn indicator demonstrating their intention to take a left turn. The cyclist was approaching in the cycle lane with a constant speed and either failed to see the indicator light of the vehicle or assumed that vehicle would yield to the cyclist. The eye tracking data showed that the driver had only one long

fixation through the left window only when they were turning the steering wheel, at which point the cyclist first appeared in the view of the driver. Both parties had to brake aggressively to prevent a collision. The proposed algorithm indicated the threat about 7 meters before the actual event.

In this instance everyone was lucky that no-one was hurt, but had the driver or the cyclist not reacted as quickly, the outcome might have been very different. Therefore, determining whether the driver has checked the mirror before initiating a turn can provide crucial fractions of a second of advance warning, potentially making the difference between a routine encounter, a near miss, a minor accident, or even a fatality. An evaluation of the effectiveness of various warning methodologies was not conducted in this study. However, it is suggested that providing an appropriate and well-timed warning to the driver could significantly enhance safety and decision-making during critical situations. Although the specific type or format of the warning is beyond the scope of this research, the findings indicate that a targeted alert system could aid drivers in reacting more promptly and appropriately to potential hazards, thereby reducing the likelihood of accidents.

IV. CONCLUSION

Driver gaze behaviour on the left wing mirror has been analysed by using eye tracking data during a left-turn manoeuvre at intersections with a cycle lane. It has been observed that many drivers fail to look at their left-wing mirror when taking left turns. Drivers are missing key visual information about the direction of their planned manoeuvre which can prove to be dangerous and could cause potential collision between vehicle and cycle lane users at intersections. An algorithm has been developed to generate an alert which can warn the driver in advance of a possible imminent collision. The algorithm is designed to be human-centric, observing the driver's gaze behaviour to raise an alert only when driver fails to look at the key gaze targets. A vehicle fitted with eye trackers and CAN bus sensors was used to carry out this study. This system could be further improved by fusing together the outwards looking sensors and cameras with eye trackers to better understand and estimate the movement of other road and cycle lane users to prevent any collision in case of missing driver gaze on key gaze targets.

In future, the driver gaze behaviour study will be expended to other road scenarios and junctions such as right-turns, lane changes, traffic merges, overtaking and T-junctions. A more advanced eye-tracking system will be incorporated to monitor the driver's gaze when they turn their head to check the area surrounding the rear of the vehicle. Other sensors including Lidar and cameras covering blind-zones could be used to improve the algorithm.

To enhance road safety, it is highly recommended by the authors of this study to develop Advanced Driver Assistance Systems (ADAS) that accurately detect cyclists and e-scooters, ensuring timely alerts without overwhelming drivers. Stakeholders, including vehicle manufacturers and regulatory bodies, should invest in these systems and prioritize driver education programs that emphasize visual attention at intersections. Urban planners should focus on infrastructure improvements that enhance the cycle lane user safety keeping

in sight the need of the urban transportation system. Ongoing research is essential for refining inattention detection algorithms, while public awareness campaigns can educate both drivers and cyclists on safe road-sharing practices. Implementing these measures will significantly reduce risks for vulnerable road users and foster a safer driving environment.

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